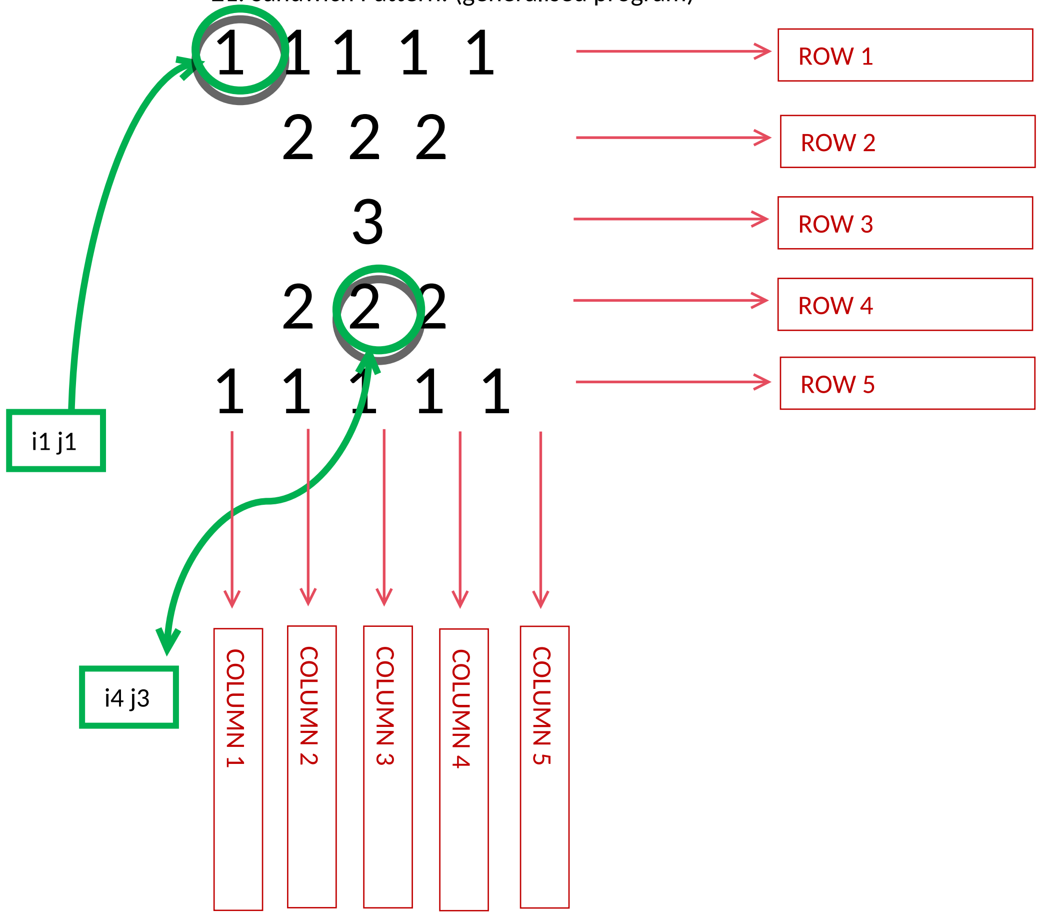
WEEK 01 MODULE
Programming Practice

TOPIC : 2 ‘ Number Pattern’

21. Sandwich Pattern: (generalised program)



1. **Problem Analysis :-**

i. Pattern Structure:

- The output forms a sandwich same number pattern.

- Unlike the Diamond Pattern, this pattern is centered from the middle and require leading spaces to shrink from middle and expand significantly giving it sandwich shape

Instead, it consists of two separate parts:

Same Number Inverted Pyramid + Same Number Pyramid

- The upper half contracts downwards

- The lower half expands downwards

ii. Execution Flow:

<u>Step 1</u>: Observe the Grid Concept:

- For n = 5

```
1 1 1 1 1
  2 2 2
    3
  2 2 2
1 1 1 1 1
```

Upper Half = Same Number Inverted Pyramid :-

```
1 1 1 1 1
  2 2 2
    3
```

Lower Half = Same Number Pyramid :-

```
  2 2 2
1 1 1 1 1
```

- Each cell in the grid represents a unique coordinate (i, j), where i denotes the row index and j denotes the column index.

-The pattern consist of two visible regions:-

✦ **Region 1: Numbers**

These shift the numbers toward the center.

For n = 5

Row 0 : []
Row 1 : [ ]
Row 2 : [ ]
Row 1 : [ ]
Row 0 : []

Relationship: Spaces = i

✦ **Region 2: Numbers**

These form the visible half diamond

For n = 5

Row 0 : [1 1 1 1 1]

Row 1 : [2 2 2]

Row 2 : [3]

Row 1 : [2 2 2]

Row 0 : [1 1 1 1 1]

Relationship : Count = 2 * (n - i) - 1 for the upper half.

- As row index increases:

Spaces increase by 1.
Numbers decrease by 2.
Printed value increases by 1.

- As row index decreases:

Spaces decrease by 1.
Numbers increase by 2.
Printed value decreases by 1.
-> Row Analysis :-

Upper Half :-

Row Index (I)	0	1	2
numbers	1	2	3
number count	5	3	1

RELATIONSHIP :-

- Stars = i + 1

Lower Half :-

Row Index (I)	1	0
number	2	1
number count	3	5

RELATIONSHIP :-

- Stars = i + 1

The formulas remain exactly the same for both halves.

Only the direction of row traversal changes.

Upper Half:

i = 0 → n - 1

Lower Half:

i = n - 2 → 0

Step 2: Traverse the grid row by row.

- The outer loop is responsible for generating rows.

Generate the upper half using the Upper Half Right Angled Triangle.

- Traverse rows from 0 to n-1.
- Print the required same number.

After the upper diamond is completed, generate the lower half.

- Traverse rows from n-2 down to 0.
- Print the required same number.

Step 3: After all numbers of the current row have been printed, move the cursor to the next line.

Step 4: Repeat the process for all rows until the outer loop terminates.

2. Condition Analysis

No conditional statements are required.

There is no need for if(...)

- Every position visited by the number loop prints a number.

- The pattern is controlled entirely through loop limits.

3. Core Logic Transformation

Same Number Inverted Pyramid Pattern:-

for(int j = 0; j < 2 * (n - i) - 1; j++)

- Meaning:

The number count decreases by 2 in every row.

Numbers move toward the center while the visible region shrinks.

Same Number Pyramid Pattern:-

for(int j = 0; j < 2 * i + 1; j++)

- Meaning:

The number count increases by 2 in every row.

Numbers expand outward from the center.

Sandwich Pattern:-

for(int i = 0; i < n; i++)

- Meaning:

Generates the shrinking upper half.

The number count decreases by 2 in every row while leading spaces increase by 1.

---

for(int i = n - 2; i >= 0; i--)

- Meaning:

Generates the expanding lower half.

The number count increases by 2 in every row while leading spaces decrease by 1.

The formula for printing numbers remains:

for(int j = 0; j < 2 * (n - i) - 1; j++)

Only the row traversal direction changes.

This creates the complete half diamond shape

4. Program :-

```
package week1_module;
import java.util.Scanner;
public class StarPattern {

        public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of rows: ");
        int n = sc.nextInt();
        // Upper Half
        for(int i = 0; i < n; i++) {
            for(int j = 0; j < i; j++) {
                System.out.print(" ");
            }
            for(int j = 0; j < 2 * (n - i) - 1; j++) {
                System.out.print((i + 1) + " ");
            }

            System.out.println();
        }

        // Lower Half
        for(int i = n - 2; i >= 0; i--) {
            for(int j = 0; j < i; j++) {
                System.out.print(" ");
            }
            for(int j = 0; j < 2 * (n - i) - 1; j++) {
                System.out.print((i + 1) + " ");
            }

            System.out.println();
        }

        sc.close();
    }
}
```

If the pattern starts from some number k, then:

Value = i + k

✦ **Why n - 2?**

Many students write:

for(int i = n - 1; i >= 0; i--)

and get:

```
1  1  1  1  1
   2  2  2
      3
      3
   2  2  2
1  1  1  1  1
```

here the Middle row is printed twice
as Upper half already printed:

3
So we start from: i = n - 2

which skips the duplicate middle row.

hence obtained figure:-

```
1  1  1  1  1
   2  2  2
      3
   2  2  2
1  1  1  1  1
```

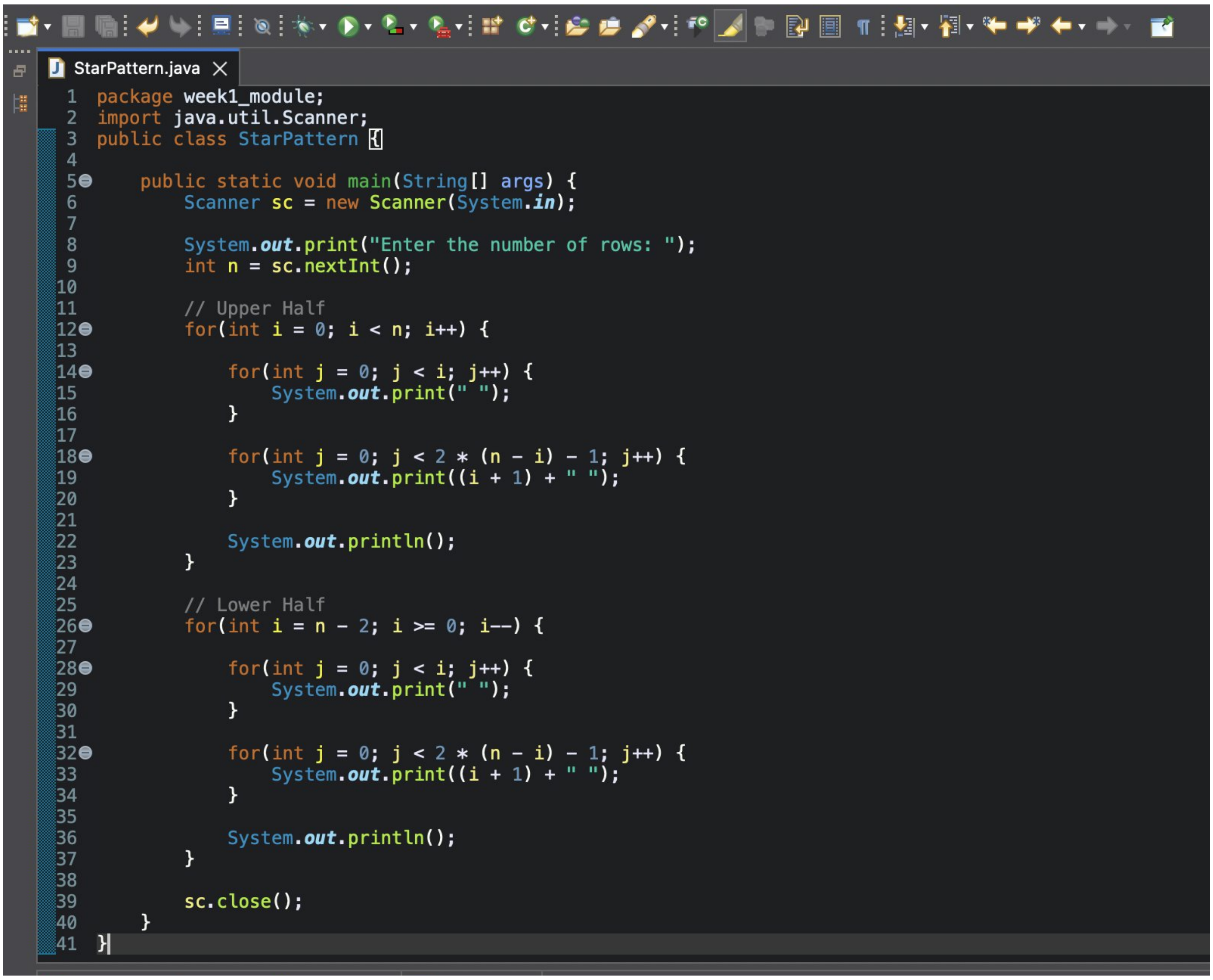
5. Program Output :-

Enter the number of rows: 5

```
1  1  1  1  1
   2  2  2
      3
   2  2  2
1  1  1  1  1
```

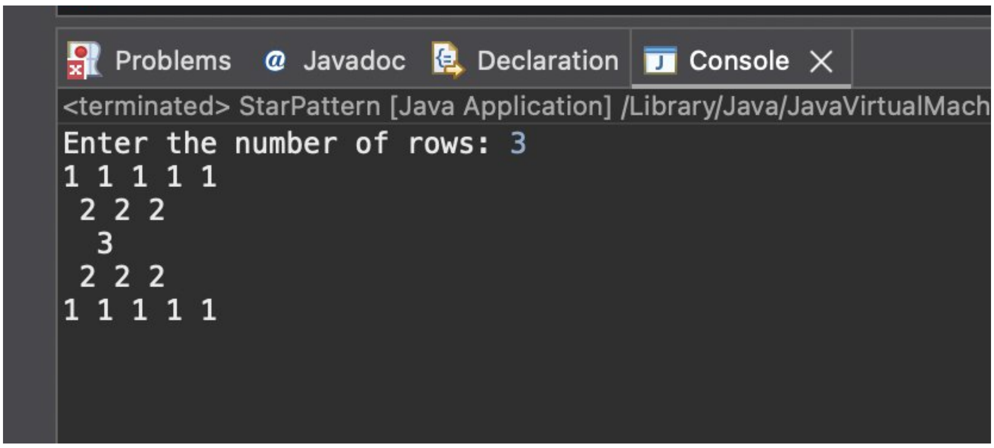
6. ScreenShot :-

i. Program :-



```java
package week1_module;
import java.util.Scanner;
public class StarPattern {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of rows: ");
        int n = sc.nextInt();

        // Upper Half
        for(int i = 0; i < n; i++) {

            for(int j = 0; j < i; j++) {
                System.out.print(" ");
            }

            for(int j = 0; j < 2 * (n - i) - 1; j++) {
                System.out.print((i + 1) + " ");
            }

            System.out.println();
        }

        // Lower Half
        for(int i = n - 2; i >= 0; i--) {

            for(int j = 0; j < i; j++) {
                System.out.print(" ");
            }

            for(int j = 0; j < 2 * (n - i) - 1; j++) {
                System.out.print((i + 1) + " ");
            }

            System.out.println();
        }

        sc.close();
    }
}
```

ii. Program's Output :-



```
Problems  @ Javadoc  Declaration  Console
<terminated> StarPattern [Java Application] /Library/Java/JavaVirtualMach
Enter the number of rows: 3
1 1 1 1 1
 2 2 2
  3
 2 2 2
1 1 1 1 1
```